Yanbo Zhou

Ph.D. Candidate
Non-Volatile Systems Lab (NVSL)
University of California, San Diego

yaz093@ucsd.edu
(858) 333-0347
https://yanbozyb.github.io/

Research Interests

My research focuses on computer systems for cloud and AI infrastructure, with a particular emphasis on memory and storage systems. My goal is to make computer systems faster, more resource-efficient, and more reliable for traditional and agentic cloud workloads at large scale.

Education

09/2023 - Present — **University of California, San Diego** — San Diego, USA
Ph.D. in Computer Science
Advisor: Prof. Steven Swanson

09/2016 - 01/2019 — **East China Normal University** — Shanghai, China
M.Eng. in Software Engineering

09/2012 - 06/2016 — **Taiyuan University of Technology** — Taiyuan, China
B.Eng. in Software Engineering

Research Experience

09/2023 - Present — **University of California, San Diego** — San Diego, USA
Graduate Student Researcher, Non-Volatile Systems Lab (NVSL)

Led research on memory and storage systems, advised by Prof. Steven Swanson.

- **Distributed Sandbox for Agents**: an agent-native distributed sandbox for fast and secure tool call and code execution (in submission).
- **Software-defined Elastic Memory**: a software-defined elastic memory system for virtual machines, which achieves flexible allocation of memory capacity and bandwidth and performance isolation (published at OSDI'26).
- **Energy-efficient Storage System**: a fast and sustainable storage system for all-flash servers with a CPU scheduler for storage software stack, which enables high performance and energy efficiency (published at SOSP'25).

01/2019 - 06/2023 — **Alibaba Cloud** — Hangzhou, China
Senior (Research) Engineer, Elastic Block Storage (EBS) Team

Led research and development of block storage systems in the cloud to address challenges of deployed systems at scale and advance cloud storage systems.

- **ASIC/SoC co-designed DPU**: a cloud-naive DPU architecture for NVMe SSD-based storage to achieve fast, compute-efficient, and flexible cloud local storage (published at FAST'26 and received **Best Paper Award**).
- **Cloud Storage Acceleration Layer**: a cache abstraction designed for high-density QLC SSDs and Zoned Namespace technology, which enabled HDD to QLC SSD transition in cloud storage systems (published at EuroSys'24).
- **Reliable Virtualized Storage Pool**: a new virtualization system to resolve storage reliability challenges due to unexpected software and hardware failures of cloud storage systems (published at USENIX ATC'20).

Publications

Conference and Journal Publications

TOS'26 **Cloud local storage goes elastic: from hardware offloading to hybrid architecture**
Leping Yang, **Yanbo Zhou**, Gong Zeng, Li Zhang, Saisai Zhang, Ruilin Wu, Chaoyang Sun, Shiyi Luo, Wenrui Li, Keqiang Niu, Xiaolu Zhang, Junping Wu, Jiaji Zhu, Jiesheng Wu, Mariusz Barczak, Wayne Gao, Ruiming Lu, Erci Xu, and Guangtao Xue
2026 ACM Transactions on Storage (TOS)

OSDI'26 **Break on Through to the Other Side: Pooling Memory Elastically with RamRyder**
Yanbo Zhou, Erci Xu, Dongjoo Seo, Adam Manzanares, Steven Swanson.
20th USENIX Symposium on Operating Systems Design and Implementation (OSDI)
[paper] [project]

FAST'26 **Here, There and Everywhere: The Past, the Present and the Future of Local Storage in Cloud**
Leping Yang, **Yanbo Zhou**, Gong Zeng, Li Zhang, Saisai Zhang, Ruilin Wu, Chaoyang Sun, Shiyi Luo, Wenrui Li, Keqiang Niu, Xiaolu Zhang, Junping Wu, Jiaji Zhu, Jiesheng Wu, Mariusz Barczak, Wayne Gao, Ruiming Lu, Erci Xu, and Guangtao Xue
24th USENIX Conference on File and Storage Technologies (FAST) [paper]
Best Paper Award

SOSP'25 **Sleeping with One Eye Open: Fast, Sustainable Storage with Sandman**
Yanbo Zhou, Erci Xu, Anisa Su, Jim Harris, Adam Manzanares, Steven Swanson.
31st ACM Symposium on Operating Systems Principles (SOSP) [paper]

EuroSys'24 **CSAL: the Next-Gen Local Disks for the Cloud**
Yanbo Zhou, Erci Xu, Li Zhang, Kapil Karkra, Mariusz Barczak, Wayne Gao, Wojciech Malikowski, Mateusz Kozlowski, Łukasz Łasek, Ruiming Lu, Feng Yang, Lilong Huang, Xiaolu Zhang, Wenrui Li, Jinhu Li, Keqiang Niu, Jiaji Zhu, Jiesheng Wu.
19th ACM European Conference on Computer Systems (EuroSys) [paper] [project]

ATC'20 **Spool: Reliable Virtualized NVMe Storage Pool in Public Cloud Infrastructure.**
Shuai Xue, Shang Zhao, Quan Chen, Gang Deng, Zheng Liu, Jie Zhang, Zhuo Song, Tao Ma, Yong Yang, **Yanbo Zhou**, Keqiang Niu, Sijie Sun, Minyi Guo.
2020 USENIX Annual Technical Conference (ATC) [paper]

RTCSA'18 **Write-aware Data Allocation on Heterogeneous Memory Architecture with Minimum Cost.**
Yanbo Zhou, Shouzhen Gu, Lixia Zheng, Edwin H.-M. Sha, Qingfeng Zhuge, Lin Wu.
24th IEEE International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA) [paper]

SC2'18 **Accelerating I/Os in virtual machines on physical NVMe SSDs via user space vhost target.**
Ziye Yang, Changpeng Liu, **Yanbo Zhou**, Xiaodong Liu, Gang Cao
8th IEEE International Symposium on Cloud and Service Computing (SC2) [paper] [project]

Preprints

2026 **SmoothAgent: Efficient Long-Horizon LLM-Based Agent Serving with Lookahead Context Engineering**
Zaifeng Pan, Qianxu Wang, Zhengding Hu, Chang Chen, Yue Guan, **Yanbo Zhou**, Steven Swanson, and Yufei Ding.
arXiv preprint arXiv:2607.00151 [paper]

Tech Reports

2023 **A Media-Aware Cloud Storage Acceleration Layer (CSAL) Cache Solution with Intel Optane SSDs for Alibaba ECS Local Disk D3C Service.**
Yanbo Zhou, Li Zhang, Kapil Karkra, Wayne Gao, Chunhong Mao, Mariusz Barczak.
Intel White Paper [paper].

Book Chapters

2019 **Linux Open Source Storage: from Ceph to Container (Chinese Edition)**
In the Publishing House of Electronics Industry [book]

Talks

Break on Through to the Other Side: Pooling Memory Elastically with RamRyder

07/2026 OSDI'26

06/2026 UCLA System Group

05/2026 SRC ACE Center for Evolvable Computing [slides] [talk]

Here, There and Everywhere: The Past, the Present and the Future of Local Storage in Cloud

04/2026 Meta AI and Systems Co-Design Team

02/2026 FAST'26 [slides] [talk]

Sleeping with One Eye Open: Fast, Sustainable Storage with Sandman

12/2025 SRC ACE Center for Evolvable Computing

09/2025 Meta AI and Systems Co-Design Team [slides]

CSAL: the Next-Gen Local Disks for the Cloud

04/2025 EuroSys'24 [slides]

Best SPDK Practices: Lessons from Five Years of Storage Evolution in Alibaba Cloud

12/2022 SPDK PRC Forum [slides] [talk - chinese]

Removing QLC Write-Amplification through Intel Optane SSD with SPDK WSR

12/2021 SPDK PRC Forum [slides] [talk - chinese]

Academic Service

Reviewer

2026 IEEE Computer Architecture Letters (CAL)

Artifact Evaluation Committee

2026 USENIX FAST, USENIX OSDI

2025 ACM ASPLOS, ACM EuroSys, ACM SOSP

Teaching Experience

Fall 2025 **Introduction to Computer Architecture: A Software Perspective (UCSD CSE 142)**
Head Teaching Assistant
Head TA for an undergraduate computer architecture course with 90+ students; led four tutors and collaborated with instructor (Prof. Steven Swanson) to design assignments, quizzes, and exams. Held office hours to help students debug and reason through technical issues related to computer architecture.